SCIENCE 1.6 **Name:**_____

WORKSHEET FIVE **FORCES AND NEWTON'S LAWS**

1. (a) What is the SI unit for mass? _____ What is the symbol for this unit? _____(A)

   (b) Write a definition for mass. (A)_____

   _____

   (c) Forces are sometimes described by the words **push, pull** and **twist.** Give an example of each of these three types of forces. (A)

   Push _____

   Pull _____

   Twist _____

2. List two advantages and two disadvantages of friction. (A)

   Advantages _____

   _____

   Disadvantages _____

   _____

3. Explain why a falling object will reach a constant speed of fall (Terminal velocity). (M)

   _____

   _____

   _____

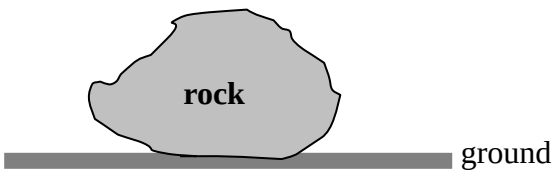
4. The forwards in a Provincial rugby team have the following masses: 120kg, 110kg, 105kg, 100kg, 98kg, 95kg, 90kg and 90kg. Calculate the total mass of the pack and also what this forward pack would weigh in Newtons. (A,M)

   _____

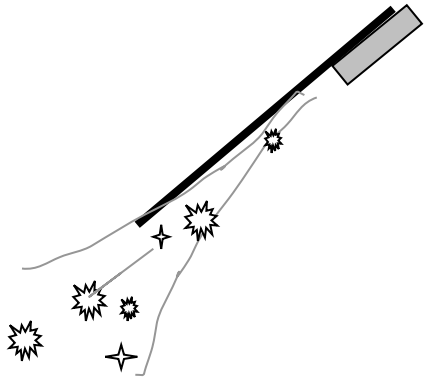
5. Write a definition for weight. (A)_____

   _____

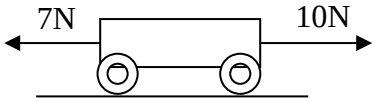
6. Draw and label all the forces acting on the rock. (A)



7. Draw and label all the forces acting on the skyrocket. (M)

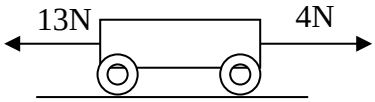


8. In the following diagrams calculate the net force on the 1 kg trolleys and also their acceleration. (A,M)

(a) 

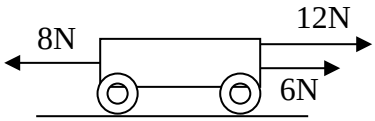
Net Force = _____

Acceleration = _____

(b) 

Net Force = _____

Acceleration = _____

(c) 

Net Force = _____

Acceleration = _____

9. **"An object will not accelerate when the forces acting on it are balanced**". Explain what this statement means. (M)

_____

_____

10. Use the equation F = ma to find the missing values in the chart below. (A)

Force (N)				800	45	24	1000
Mass (kg)	100	2.5	0.3	80	5		
Acceleration (ms^{-2})	3	20	15			48	200